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RESEARCH

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Effects of home- and gym-based resistance training on glycemic control in patients with type 2 diabetes mellitus—a systematic review and meta-analysis

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Abstract

Background Targeting skeletal muscle mass and quality through resistance training (RT) may be a particularly promising approach for treating metabolic diseases. While the benefits of RT in managing type 2 diabetes mellitus (T2DM) are well established, most studies supporting these benefits have been conducted in controlled laboratory or gym settings. To date, it remains unclear whether home-based RT can be as effective as gym-based RT in improving the glycemic profile of patients with T2DM. Therefore, the primary aim of this meta-analysis was to compare the effects of home-based versus gym-based RT on glycemic control (HbA1c) in patients with T2DM.

Methods A systematic literature search was conducted using SPORTDiscus, PubMed and BISp SURF databases (updated until 2 August 2024). After screening, 20 controlled trials (involving 1397 participants) met the inclusion criteria.

Results A random-effects model revealed a significant effect of RT on glycemic control, with a pooled mean difference favoring RT versus control (mean difference: −0.33; 95% confidence interval (CI): −0.49 to −0.18). Gym-based RT significantly reduced HbA1c compared to control conditions (−0.39; 95% CI: −0.57 to −0.22), while home-based RT showed no significant effect (+0.12; 95% CI: −0.16 to +0.39). Heterogeneity was substantial**,** suggesting considerable variability between studies. The methodological quality score of the included studies, assessed using the PEDro (Physiotherapy Evidence Database) scale, ranged from 3 to 8, with an average score of 6 ± 1 ("good" quality).

Conclusions This meta-analysis confirms the effectiveness of RT in reducing HbA1c levels in individuals with T2DM. Notably, the present findings highlight that gym-based RT is effective, while home-based RT is not. Possible reasons include increased motivation by coaches or training buddies in the gym, resulting in increased adherence to the training program, as well as the limited availability of equipment or imprecise load dosing during home workouts.

Review registration: CRD420250650823.

Keywords Strength, Resistance, Exercise, Training, Physical activity, Diabetes, HbA1c

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Introduction

Type-2-diabetes mellitus (T2DM) is a major global health concern, contributing to rising morbidity and mortality worldwide [1]. Its progression is associated with a range of comorbidities, including microvascular and macrovascular complications and mental disorders [2, 3]. The global prevalence of T2DM has been steadily increasing over the last three decades [4]. Developing effective strategies for the prevention and treatment of T2DM are urgently needed. Physical activity and tailored exercise interventions are well-established strategies for improving insulin sensitivity, regulating blood glucose levels, and reducing the risk of diabetes-related complications [5, 6].

Skeletal muscle plays a crucial role in regulating glucose homeostasis, accounting for up to ~80% of insulin-stimulated glucose uptake in the postprandial state. However, in individuals with T2DM, glucose uptake is significantly impaired due to insulin resistance [7]. Several mechanisms, including oxidative stress and inflammation, likely contribute to this dysfunction [8].

Targeting skeletal muscle mass and quality through resistance training (RT) may be a particularly promising approach for managing T2DM. Previous meta-analyses have demonstrated the efficacy of RT in improving glycemic control [9–14]. While the exact mechanisms underlying RT-induced improvements in glycemic control are not yet fully understood, they may involve increased muscle mass as well as increased availability of insulin-signaling pathway molecules, alongside an enhanced anti-oxidative capacity and anti-inflammatory protection system [7].

While the benefits of RT in managing T2DM are widely accepted, most studies supporting these benefits have been conducted in controlled laboratory or gym settings. To date, it remains unclear whether home-based RT can be as effective as gym-based RT in improving the glycemic profile of patients with T2DM.

Home-based RT offers several potential advantages compared to RT in gym settings. Engaging in RT at home can help individuals with T2DM overcome commonly reported barriers such as limited access to exercise facilities [15–18]. It offers a cost-effective alternative, requiring minimal equipment while eliminating expenses related to gym memberships and transportation. Moreover, home-based training provides greater flexibility in scheduling training, which is particularly important since lack of time is another frequently stated barrier among individuals with T2DM [19–21]. Engaging in RT at home may furthermore help reduce psychological barriers such as feelings of shame, which are commonly reported by individuals with T2DM and obesity [22]. On the other hand, gym-based training has its own advantages. Exercise professionals play a key role in promoting exercise engagement among T2DM patients by increasing awareness of the benefits of physical activity and providing social support, which can increase patients' motivation [18]. Furthermore, supervised training ensures the exercises are properly executed and performance is monitored. In addition, patients in the home setting are presumably more often left to their own devices to progressively increase the workload to achieve improved training performance.

Therefore, the primary aim of this systematic analysis was to compare the effects of home-based RT and gym-based RT on glycemic control in patients with T2DM to address a gap in existing literature, particularly on the efficacy of home-based resistance training. We also evaluated whether training duration has a considerable effect by comparing long-term RT (> 4 months) with short-term RT (≤ 4 months). In addition, the role of different strength training methods was explored. Thereby, this meta-analysis aims to update the evidence on resistance training for patients with T2DM.

Methods

Search strategy

The review has been registered in PROSPERO (CRD420250650823). A literature search was conducted following PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) guidelines [23] on PubMed, SPORTDiscus and BISp SURF databases (original search: May 2024, updated until 2 August 2024). The full search strings are available in the appendix (Additional file 1).

Eligibility criteria

Eligibility criteria were defined using the Patient-Intervention-Comparison-Outcome (PICO) scheme. Only randomized controlled trials (RCTs) published in peer-reviewed journals in English were included for further analysis. Only interventions involving adult patients (≥ 18 years, all sexes, diagnosed with T2DM and/or HbA1c ≥ 6.5%) were included. Taking medication, including insulin, was not defined as an exclusion criterion (P). Moreover, only trials with at least one resistance training intervention were considered. The minimum information on the resistance training program was: gym-based (including training sessions at research institutes) or home-based (with clearly defined periods during which either only gym-based or home-based training was done, but not a mix of both), supervised/unsupervised, duration of the intervention, details on the training (e.g. number of sets, repetitions per set, training intensity) that allow an accurate identification of the training method used (strength endurance training, hypertrophy training,

maximum strength training). Aerobic endurance and stretching exercises were permitted only during the warm-up or cool-down phases (I). The minimum intervention duration was eight weeks. Studies also had to include a control group (receiving usual care or the same care as the intervention group but without a structured, progressive exercise program) (C). Finally, only studies evaluating glycemic control (HbA1c) were considered (O).

Study selection and data collection

The retrieved records were imported into *Rayyan* (https://www.rayyan.ai) [24]. All records were screened by title and abstract after removing duplicates. Full-text articles were screened in the next step. Two researchers (MB and SG) independently assessed the articles for eligibility. Any discrepancies were discussed with a third researcher (CB) until consensus was reached.

The extracted data included the first author and year of publication, country, number of participants (including sex and age), study design, intervention duration, training frequency, volume and intensity of single sessions, exercises and equipment used as well as the pre- and post-training HbA1c levels. If not reported in the papers, change-from-baseline values were calculated by subtracting the $mean_{pre}$ value from the $mean_{post}$ value ($\Delta = mean_{post} - mean_{pre}$) for each group (experimental and control group). If missing change-from-baseline standard deviations (SD) could not be calculated from the study's data, they were imputed using SDs from other studies with a similar study duration following the standards of the Cochrane Handbook 5.1. (https://handbook-5-1.cochrane.org, chapter 16.1.3.2 "Imputing standard deviations for changes from baseline" [25]). Where available, data were extracted from per protocol analyses (16 studies) (otherwise from intention-to-treat analyses (4 studies)).

Data analyses

A random-effects model was used to calculate the pooled mean difference (MD) and corresponding 95% confidence interval (CI) as a measure of treatment effectiveness [26]. Heterogeneity was assessed using the I^2 statistic and interpreted as low (≤ 50%), moderate (50–75%) or high (> 75%) heterogeneity. To evaluate the robustness of the findings, a leave-one-out analysis was performed, sequentially omitting each study and recalculating the pooled effect size and confidence interval. Publication bias was assessed using funnel plots and Egger's regression test [27].

Subgroup analyses were conducted to explore potential moderating factors, including the training setting (gym-based versus home-based), type of resistance training, and intervention duration. Resistance training was categorized as strength endurance training (most of the time: 15 + reps, approx. < 60% maximal power/1-repetition maximum (1-RM)), hypertrophy training (8–15 reps, approx. 60–80% maximal power/1-RM), as well as maximal strength training (1–5 reps, > 80% maximal power/1-RM) [28]. Studies were further categorized into short (≤ 4 months) and long-term interventions (> 4 months) based on the median intervention duration across the included studies.

Additionally, a meta-regression analysis was conducted to examine the relationship between the total number of training sessions completed and the HbA1c MD. This analysis assessed whether greater training exposure was associated with larger changes in HbA1c levels. All statistical analyses were performed using R software (version 4.1.1; The R Foundation for Statistical Computing) with the'meta'package [29].

Quality assessment of randomized controlled trials and evaluation of the certainty of evidence supporting recommendations for gym-based and home-based resistance training

The methodological quality and risk of bias were assessed by MG and VI using the PEDro (Physiotherapy Evidence Database) scale [30]. In case of disagreement on the rating of an item, a third rater (CB) made the final decision. The PEDro scale consists of 11 items, in which Criteria 2–9 assess randomization and internal validity, while Criteria 10–11 assess the presence of statistically replicable results. Criterion 1 relates to external validity but was not included in the calculation of the final PEDro score. A higher score indicates a better study quality [30].

As blinding of instructors and participants in exercise intervention studies is difficult but not entirely impossible by using sham interventions, items 5–6 were rated accordingly. Scores of 3 or less indicate "poor", scores of 4–5 "fair", scores of 6–8 "good", and scores of 9–10 "excellent" quality. The certainty of evidence was evaluated using the GRADE (Grading of Recommendations, Assessment, Development and Evaluation) approach, which has been used in other meta-analyses of physical activity interventions [31]. Factors such as study limitations (risk of bias), inconsistency, indirectness, imprecision, and publication bias were evaluated, resulting in the downgrading or upgrading of the certainty of evidence. The GRADE system categorizes the quality of evidence into four levels: high, moderate, low, and very low. Any discrepancies were discussed between the authors until consensus was reached.

Results

Literature search

The initial search yielded a total of 413 records. After removing duplicates, 409 unique records remained. These were screened based on their titles and abstracts, resulting in the exclusion of 369 studies that did not meet the inclusion criteria. A total of 40 studies were considered potentially eligible for full-text review. After a detailed evaluation of the full texts, 20 studies were excluded due to reasons such as inappropriate study design or insufficient outcome data (a detailed list of reasons for exclusion is available in the appendix (Additional file 2)). Ultimately, 20 studies met the inclusion criteria and were included in the meta-analysis. Figure 1 presents the PRISMA flow diagram summarizing the study selection process.

Study characteristics

The included studies comprised a total of 1,397 participants in the present meta-analysis, with sample sizes

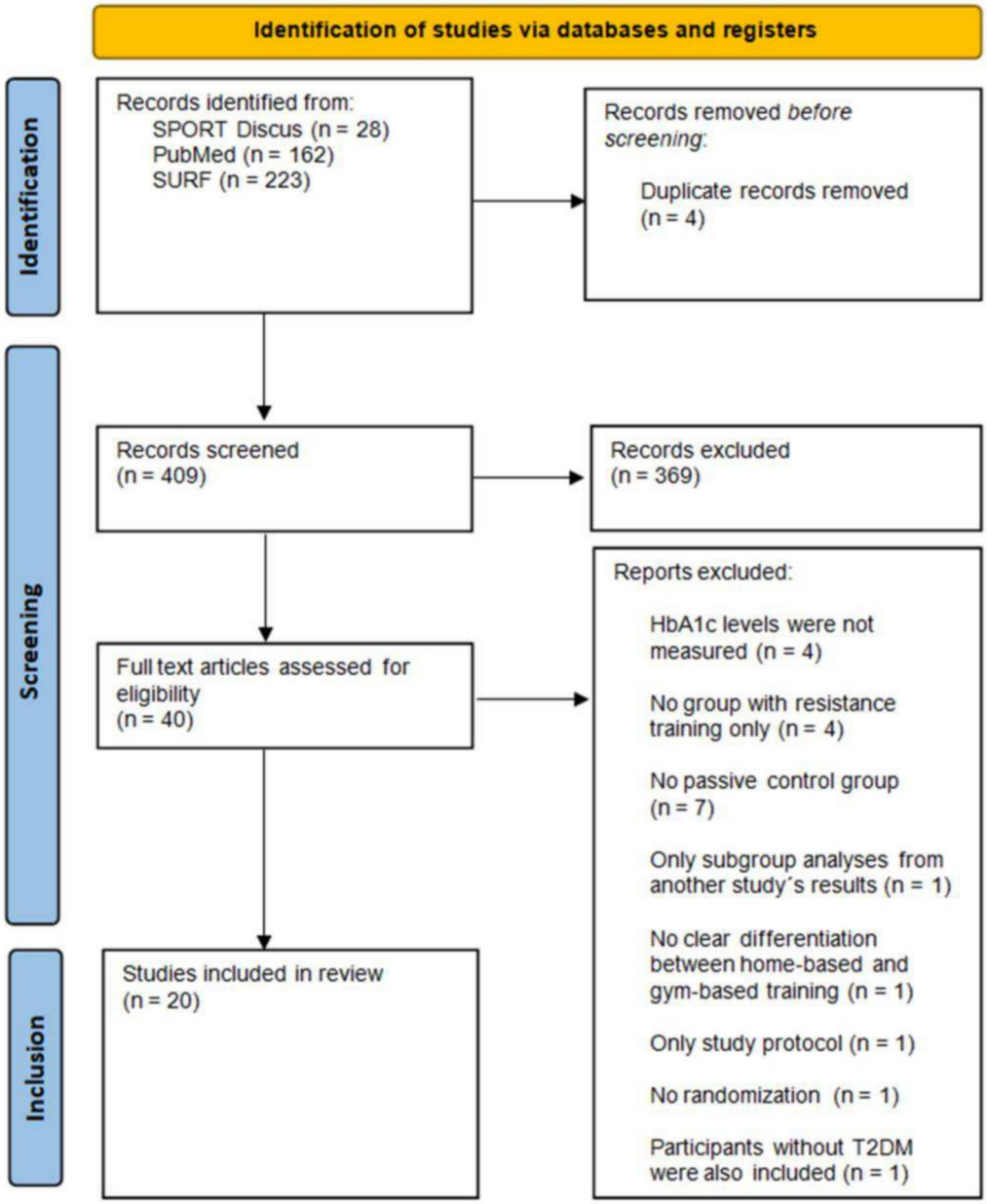

Fig. 1 PRISMA flow chart

ranging from 20 to 127 and an average of 64 ± 55 participants per study (considering only the resistance and control groups). One study included only men [32], while another included only women [33]. All other studies included both males and females. All studies only involved patients diagnosed with T2DM. The presence of overweight/obesity was explicitly mentioned as an inclusion criterion in two studies, while very obese patients were excluded in three studies. The participants' mean ages ranged from 48 to 68 years. The duration of the resistance training interventions varied between 8 and 52 weeks, with an average of 26 ± 14 weeks. Resistance training protocols included various approaches, such as maximal strength training, hypertrophy training, and strength endurance training, focusing on HbA1c outcomes. Detailed descriptions of the population groups and training protocols are provided in Table 1.

Meta-analytic data

Gym-based and home-based resistance training to improve glycemic control

The forest plot presents the individual study estimates and the overall effect size for the impact of resistance training on HbA1c reduction (Fig. 2). The random-effects meta-analysis yielded a statistically significant but small pooled mean difference of −0.33% [95% CI − 0.49 to − 0.18] in favor of resistance training. However, the heterogeneity was substantial (I^2 = 74.0%, p < 0.0001), indicating considerable variability between studies. A leave-one-out sensitivity analysis showed that the pooled effect size remained stable when individual studies were excluded, with mean differences ranging from −0.36% to −0.30%. This indicates that no single study—including those with lower methodological quality—had a relevant influence on the overall result.

Subsequent subgroup analyses revealed notable differences between home-based and gym-based resistance training modalities (Fig. 3). Gym-based resistance training showed a small but significant effect versus control conditions (− 0.39% [95% CI − 0.57 to − 0.22]), while home-based training showed no significant effect (+ 0.12% [95% CI − 0.16 to + 0.39]). Furthermore, heterogeneity was high in the gym-based subgroup (I^2 = 78.6%), but negligible in the home-based subgroup (I^2 = 0%). A test for subgroup differences confirmed a statistically significant difference (p = 0.0023) between the two modalities, suggesting that gym-based training may be more effective in reducing HbA1c levels than home-based training.

Identification of potential factors influencing the effects of gym-based and home-based resistance training on glycemic control

A subgroup analysis based on intervention duration (Fig. 4) revealed no clear advantages for longer interventions (> 4 months) than for shorter ones (≤ 4 months). Studies with shorter interventions demonstrated a small to moderate pooled mean difference (− 0.54% [95% CI − 0.91 to − 0.18]), whereas longer interventions showed a smaller, but still significant effect (− 0.26% [95% CI − 0.42 to − 0.10]). Both subgroups favored resistance training, although heterogeneity was higher in the longer intervention subgroup (I^2 = 76.7%) than in the shorter one (I^2 = 54.5%). The test for subgroup differences was not statistically significant (p = 0.1661), indicating that intervention duration did not substantially moderate the effects of resistance training on HbA1c reductions.

Furthermore, the meta-regression analysis revealed no significant association between the total number of training sessions and the reduction in HbA1c levels (β = 0.001, SE = 0.002, p = 0.609, Fig. 5). This suggests that the total volume of resistance training alone may not be a primary determinant of HbA1c improvement. The analysis showed substantial heterogeneity among studies (I^2 = 80.7%), indicating considerable variability in study designs and participant characteristics. Additionally, the proportion of variance explained by the regression model was very low (R^2 = 0%), suggesting that the number of training sessions does not account for the observed differences in HbA1c outcomes.

The subgroup analysis of different resistance training methods (Fig. 6) revealed variations in effectiveness. Hypertrophy training yielded a significant pooled mean difference (− 0.39% [95% CI − 0.56 to − 0.23]), while strength endurance training (− 0.02% [95% CI − 0.38 to + 0.33]) and maximal strength (− 0.01% [95% CI − 0.18 to + 0.16]) training did not show significant effects. Heterogeneity was high in the hypertrophy training subgroup (I^2 = 76.9%), whereas it was minimal (I^2 = 0.0%) in the strength endurance and maximal strength training groups. The test for subgroup differences was statistically significant (p = 0.0041), suggesting that the training method used may influence the effectiveness of resistance training on HbA1c reduction. However, only two studies on strength endurance training and one on maximal strength training were included in the meta-analysis.

Finally, the funnel plot analysis (Fig. 7) did not reveal relevant indications of publication bias, suggesting that the results are not substantially influenced by systematic reporting biases.

Table 1 Study characteristics of included randomized-controlled trials

Study	Country	Total number of participants (of all groups)	Mean age (± SD, unless otherwise stated)	Number of participants in the resistance training and control groups ultimately considered for analysis	Duration	Frequency[b]	Volume and intensity[b]	Exercises, equipment and duration of a single session[b]	PEDro score
Al Ozairi 2023 [34]	Kuwait	• Total (N) = 120 • M (n) = 74 • F (n) = 46	60 (9)	• RT: 26 • C: 36	32 wks	3/wk	• Wk 1: 1 set × 5–10 reps that were tiring but comfortably achievable • Wk 2–4: 1 set × 15–20 reps performed up to voluntary muscular failure • Wk 5–8: 2 sets × 15–20 reps performed up to voluntary muscular failure • Wk 9–32: 3 sets × 15–20 reps performed up to voluntary muscular failure	• Exercises: press ups, band lateral raises, band seated low row, squat, lunge, calf raise and plank • Equipment: not reported • Duration of session: not reported	6
Arora 2009 [35]	India	• Total (N) = 30 • M (n) = 16 • F (n) = 14	54 (9)	• RT: 9 • C: 10	8 wks	2/wk	• 3 sets × 10 reps • Intensity: progression from 60% of 1RM to 100% of initial 1RM from wk 1 to wk 8	• 7 exercises for major muscle groups: biceps, triceps, upper back, abdominals, knee flexors and extensors • Equipment: dumbbells, pulleys, lateral pull down and quadriceps table • Duration of session: not reported	5

Table 1 (continued)

Study	Country	Total number of participants (of all groups)	Mean age (± SD, unless otherwise stated)	Number of participants in the resistance training and control groups ultimately considered for analysis	Duration	Frequency[b]	Volume and intensity[b]	Exercises, equipment and duration of a single session[b]	PEDro score
Baldi 2003 [32]	New Zealand	• Total (N) = 18 • M (n) = 18	48	• RT: 9 • C: 9	10 wks	3/wk	• Wk 1: 1 set x 10RM (upper body) and 15RM (lower body) • Wk 2–10: 2 sets × 12 reps at initial 10RM (upper body) and 15RM (lower body), increasing 5% of weight when subject successfully completed the sets and reps	• 10 exercises involving the major muscle groups of the upper and lower body • Equipment: not reported • Duration of session: not reported	3
Brooks 2006 [36]	USA	• Total (N) = 62 • M (n) = 40 • F (n) = 22	• RT = 66 (SE: 2) • C = 66 (SE: 1)	• RT: 31 • C: 31 All patients, ITT analysis	16 wks	3/wk	• Wk 1–8: 3 sets × 8 reps at 60–80% of baseline 1RM • Wk 9–10: 3 sets × 8 reps at 70–80% of mid-study 1RM	• Exercises: upper back, chest press, leg press, knee extension and flexion • Equipment: pneumatic machines • Duration of session: 45 min	7
Castaneda 2002 [37]	USA	• Total (N) = 62 • M (n) = 22 • F (n) = 40	66 (SE: 8)	• RT: 31 • C: 31	16 wks	3/wk	• Wk 1–8: 3 sets × 8 reps at 60–80% of baseline 1RM • Wk 9: 3 sets × 8 reps at 10% lower than current workload • Wk 10–14: 3 sets × 8 reps at 70–80% of mid-study 1RM • Wk 15: 3 sets × 8 reps at 10% lower than current workload	• Exercises: upper back, chest press, leg press, knee extension and flexion • Equipment: pneumatic machine • Duration of session: 45 min	6

Table 1 (continued)

Study	Country	Total number of participants (of all groups)	Mean age (± SD, unless otherwise stated)	Number of participants in the resistance training and control groups ultimately considered for analysis	Duration	Frequency[b]	Volume and intensity[b]	Exercises, equipment and duration of a single session[b]	PEDro score
Chien 2022 [38]	Taiwan	• Total (N) = 40 • M (n) = 7 • F (n) = 33	• RT = 68 (8) • C = 67 (6)	• RT: 19 • C: 18	12 wks	3/wk	Progressive increase from 3 sets × 8–15 reps at 13 RPE (scale 6–20) to 3 sets × 20 reps	• Exercises: arm curl, overhead press, hip adduction/abduction, step and tiptoe • Equipment: sandbags • Duration of session: 30 min	5
Church 2010 [39]	USA	• Total (N) = 262 • M (n) = 97 • F (n) = 165	56 (9)	• RT: 60 • C: 41	9 mos	3/wk	• 2 sets × 10–12 reps of 4 upper body exercises • 3 sets × 10–12 reps of 3 leg exercises • 2 sets × 10–12 reps of core exercises • Intensity: once the participant was able to complete 12 repetitions for each set of exercises during two consecutive exercise sessions, the prescribed weight was increased	• Exercises: bench press, seated row, shoulder press and pull down, leg press, extension and flexion, abdominal crunches and back extensions • Equipment: not reported • Duration of session: not reported	7

Table 1 (continued)

Study	Country	Total number of participants (of all groups)	Mean age (± SD, unless otherwise stated)	Number of participants in the resistance training and control groups ultimately considered for analysis	Duration	Frequency[b]	Volume and intensity[b]	Exercises, equipment and duration of a single session[b]	PEDro score
Dunstan 2005 [40]	Australia	• Total (N) = 36 • M (n) = 21 • F (n) = 15	• RT = 68 (5) • C = 67 (5)[a]	a) Gym-based training: • RT: 16 • C: 13 b) Home-based training: • RT: 14 • CT: 12	12 mos	3/wk	• Mos 1–6 gym-based training: o Wk 1–2: intensity 50–60% of 1RM o From wk 3: 3 sets × 8–10 reps at 75–85% of 1RM (except abdominal crunch) • Mos 7–12 home-based training: 3 sets × 8–10 reps at 60–80% of 1RM	• 1–6 mos: gym-based training: o Exercises: bench press, leg extension, upright row, lateral pull-down, standing leg curl (ankle weights), dumbbell seated shoulder press, dumbbell seated biceps curl, dumbbell triceps kickback and abdominal curls o Equipment: free weights and a multiple-station weight machine o Duration of session: 55 min • 7–12 mos: home-based training: o Exercises: lying dumbbell flies, seated single-leg extension (ankle weights), dumbbell shoulder press, dumbbell bent-over row, standing leg curl (ankle weights), dumbbell upright row, dumbbell bicep curls, dumbbell triceps kickbacks and abdominal curls o Equipment: dumbbells/ankle weights o Duration of session: not reported	4

Table 1 (continued)

Study	Country	Total number of participants (of all groups)	Mean age (± SD, unless otherwise stated)	Number of participants in the resistance training and control groups ultimately considered for analysis	Duration	Frequency[b]	Volume and intensity[b]	Exercises, equipment and duration of a single session[b]	PEDro score
Giessing 2022 [41]	Germany	• Total (N) = 60 • M (n) = 40[a] • F (n) = 17[a]	62 (IQR: 53)[a]	• RT: 29 • C: 28	6 mos	2/wk	• 1 set × 8–12 reps at their self-determined repetitions maximum. Upon reaching this point, participants rested for between 10–29 s and then attempted further repetitions until reaching their sdRM. This was then repeated one more time • Tempo: 2:1:2	• Exercises: chest press, seated row, reverse fly, shoulder press, elbow flexion, elbow extension, knee extension, knee flexion, lumbar extension and abdominal flexion • Equipment: Ergofit power line 4000 resistance machines • Duration of session: 30–45 min	5
Hangping 2019 [42]	China	• Total (N) = 300 • M (n) = 122[a] • F (n) = 143[a]	• RT = 66 (9) • C = 67 (7)[a]	• RT: 165 • C: 100	6 mos	1/wk	1 × maximal 5 s contractions	• Exercises: chest press, leg press, core pull and vertical lift • Equipment: Bio-Density™ • Duration of session: 5–10 min	5
Kadoglou 2013 [43]	UK	• Total (N) = 100 • M (n) = 25[a] • F (n) = 65[a]	• RT = 56 (5) • C = 58 (7)[a]	• RT: 23 • C: 24	6 mos	4/wk	• 2–3 sets × 8–10 reps at 60–80% of 1RM • 10 min calisthenics	• Exercises: seated leg press, knee extension, knee flexion, chest press, lat. pulldown, overhead press, biceps curl and triceps extension • Equipment: not reported • Duration of session: progressive increase to 60 min in the first 4 weeks and then maintained	5

Table 1 (continued)

Study	Country	Total number of participants (of all groups)	Mean age (± SD, unless otherwise stated)	Number of participants in the resistance training and control groups ultimately considered for analysis	Duration	Frequency[b]	Volume and intensity[b]	Exercises, equipment and duration of a single session[b]	PEDro score
Larose 2011 [44]	Canada	• Total (N) = 251 • M (n) = 160 • F (n) = 91	• RT = 55 (SE: 8) • C = 55 (SE: 7)	• RT: 64 • C: 63	6 mos	• 1–4 wk(run-in-phase): 2/wk • 5–26 wk (intervention): 3/wk	• Wk 1–4 (run-in-phase): 1–2 sets of 7 resistance exercises • Wk 5–26 (intervention): 3 sets × 8 reps • Intensity: when the participant could perform more than 8 reps while maintaining proper form, the weight or resistance of the exercise was increased by 2.3 to 4.5 kg	• Exercises: alternating between two groups of exercises: o Group A: abdominal crunch, seated row, biceps curl, bench press, leg press, shoulder press and leg extension o Group B: abdominal crunch, lateral pulldown, triceps pushdown, chest press, leg press, upright row and leg curl • Equipment: weight machines • Wk 5–26 duration of session: 45 min	5

Table 1 (continued)

Study	Country	Total number of participants (of all groups)	Mean age (± SD, unless otherwise stated)	Number of participants in the resistance training and control groups ultimately considered for analysis	Duration	Frequency[b]	Volume and intensity[b]	Exercises, equipment and duration of a single session[b]	PEDro score
Ma 2024 [45]	China	• Total (N) = 139 • M (n) = 39[a] • F (n) = 59[a]	66 (5)[a]	• RT: 31 • C: 33	6 mos	3/wk	• Wk 1 (pre-intervention): 2–4 sets × 8–12 reps at 40–50% of 1RM or 9–11 on RPE scale • Wk 2 (pre-intervention): 2–4 sets × 10–15 reps at 45–55% of 1RM or 9–12 on RPE scale • Wk 1–2 (intervention): 2–4 sets × 8–12 reps at 55–60% of 1RM or 12–13 on RPE scale • Wk 3–4: 2–4 sets × 10–15 reps at 55–60% of 1RM or 12–13 on RPE scale • Wk 5–8: 2–4 sets × 8–12 reps at 60–65% of 1RM or 12–13 on RPE scale • Wk 9–12: 2–4 sets × 10–15 reps at 60–65% of 1RM or 12–14 on RPE scale • Wk 13–16: 2–4 sets × 8–12 reps at 60–70% of 1RM or 12–14 on RPE scale • Wk 17–24: 2–4 sets × 10–15 reps at 60–70% of 1RM or 12–14 on RPE scale	• Exercises: shoulder press and pull down, elbow extension and flexion, leg press, extension and flexion, flat-ground support, glute bridge, push-ups and sit-ups • Equipment: barbells, dumbbells, elastic bands/ropes and kettle bells • Duration of session: 50 min	5

Table 1 (continued)

Study	Country	Total number of participants (of all groups)	Mean age (± SD, unless otherwise stated)	Number of participants in the resistance training and control groups ultimately considered for analysis	Duration	Frequency[b]	Volume and intensity[b]	Exercises, equipment and duration of a single session[b]	PEDro score
Mavros 2013 [46]	Australia	• Total (N) = 100 • M (n) = 50 • F (n) = 50	68 (6)	• RT: 36 • C: 48	12 mos	3/wk	• 3 sets × 8 reps, except 2 sets × 8 reps on each leg for hip flexion, hip extension and hip abduction at 80% of 1RM or 15–18 RPE scale • Concentric phase as fast as possible and eccentric phase in 4 secs	• Exercises: seated row, chest press, leg press, knee extension, hip flexion, hip extension and hip abduction • Equipment: pneumatic resistance equipment • Duration of session: not reported	8
Plotnikoff 2010 [47]	Canada	• Total (N) = 48 • M (n) = 16 • F (n) = 32	• RT = 55 (12) • C = 54 (12)	• RT: 27 • C: 21	16 wks	3/wk	• Wk 1: 2 sets × 10–12 reps at 50–60% of 1RM • Wk 2: 3 sets × 10–12 at 50–60% of 1RM • Wk 3–8: progressive increase to 3 sets × 10–12 reps at 70–80% of 1RM • Wk 9: 2 sets × 8–10 reps at 70% of 1RM • Wk 10–15: 3 sets × 8–10 at 70–85% of 1RM • Wk 16: 2 sets × 8–10 reps at 80% of 1RM	• Exercises: squats, seated row, chest press and shoulder press, lunges, lateral pull-down, standing triceps extension, standing pulley abdominal twists, biceps curl, triceps press, reverse rhomboid flies, lateral pulley deltoid raise and pulley abdominal curls • Equipment: Multigym apparatus and dumbbells • Duration of session: not reported	8
Ranasinghe 2021 [48]	Sri Lanka	• Total (N) = 106 • M (n) = 40[a] • F (n) = 46[a]	50 (9)[a]	• RT: 28 • C: 30	12 wks	2/wk	• 3 sets × 8 reps • Intensity: progressed according to the individual's capacity/was increased from 5% of the previous intensity level every 2 wks while maintaining 3 sets × 8 reps	• 7 exercises for major muscle groups (upper, lower body and core) • Equipment: body resistance, free weights and machines • Duration of session: 60–75 min	5

Table 1 (continued)

Study	Country	Total number of participants (of all groups)	Mean age (± SD, unless otherwise stated)	Number of participants in the resistance training and control groups ultimately considered for analysis	Duration	Frequency[b]	Volume and intensity[b]	Exercises, equipment and duration of a single session[b]	PEDro score
Samadpour 2021 [33]	Iran	• Total (N) = 40 • F (n) = 40	53 (5)	• RT: 10 • C: 10	8 wks	3/wk	• Wk 1–2: 12 reps at 14 on RPE scale • Wk 3–4: 12 reps at 14–15 on RPE scale • Wk 5–6: 10 reps at 15–16 on RPE scale • Wk 7–8: 8 reps at 17–18 on RPE scale	• Exercises: push-ups, chest press, standing rowing, lunges and squats • Equipment: TRX equipment • Duration of session: 60 min	6
Sigal 2007 [49]	Canada	• Total (N) = 251 • M (n) = 160 • F (n) = 91	• RT = 55 (8) • C = 55 (7)	• RT: 64 • C: 63	26 wks	• Wk 1–4 (run-in-phase): 2/wk • Wk 5–26 (intervention): 3/wk	• Wk 1–2 (run-in-phase): 1 set × 15 reps at 15RM • Wk 3–4 (run-in-phase): 2 sets × 15 reps at 15RM • Wk 5–10 (intervention): 3 sets × 12 reps at 12RM • Wk 11–12: 3 sets × 10 reps at 10RM • Wk 13–26: 3 sets × 8 reps at 8RM	• Exercises: alternating between two groups of exercises: o Group A: abdominal crunch, seated row, seated biceps curl, supine bench press, leg press, shoulder press and leg extension o Group B: abdominal crunch, lateral pulldown, triceps pushdown, sitting chest press, leg press, upright row and leg curl • Equipment: weight machines • Duration of session: not reported	8

Table 1 (continued)

Study	Country	Total number of participants (of all groups)	Mean age (± SD, unless otherwise stated)	Number of participants in the resistance training and control groups ultimately considered for analysis	Duration	Frequency[b]	Volume and intensity[b]	Exercises, equipment and duration of a single session[b]	PEDro score
Sparks 2013 [50]	USA	• Total (N) = 52 • M (n) = 23 • F (n) = 29	58 (8)	• RT: 18 • C: 10	9 mos	3/wk	• 2 sets × 10–12 reps of 4 upper body exercises • 3 sets × 10–12 reps of 3 leg exercises • 2 sets × 10–12 reps of abdominal exercises	• Exercises: bench press, seated row, shoulder press and lat. pull down, leg press, extension, flexion, abdominal crunches and back extensions • Equipment: not reported • Duration of session: 45–50 min	5
Yavari 2012 [51]	Iran	• Total (N) = 80 • M (n) = 37 • F (n) = 43	51 (9)[a]	• RT: 15 • C: 15	12 mos	• Mo 1: 2/wk • Mo 2–12: 3/wk	• Wk 1–2: 1–2 sets × 8–10 reps at 60% of 1RM • Wk 2–3: 1–2 sets × 8–10 reps at 75–80% of 1RM • After first month: 3 × 8–10 reps at 75–80% of 1RM	• Exercises: bench press, seated row, shoulder press, chest press, lateral pulldown, abdominal crunches, leg press, leg extension, triceps pushdown and seated bicep curls • Equipment: weight machines • Duration of session: not reported	4

C control; *F* females; *ITT* intention-to-treat, *M* males, *RM* repetition maximum, *RT* resistance training, *PEDro scale* Physiotherapy Evidence Database scale

[a] Information refers to those who completed the study (study finishers)

[b] Information refers exclusively to RT

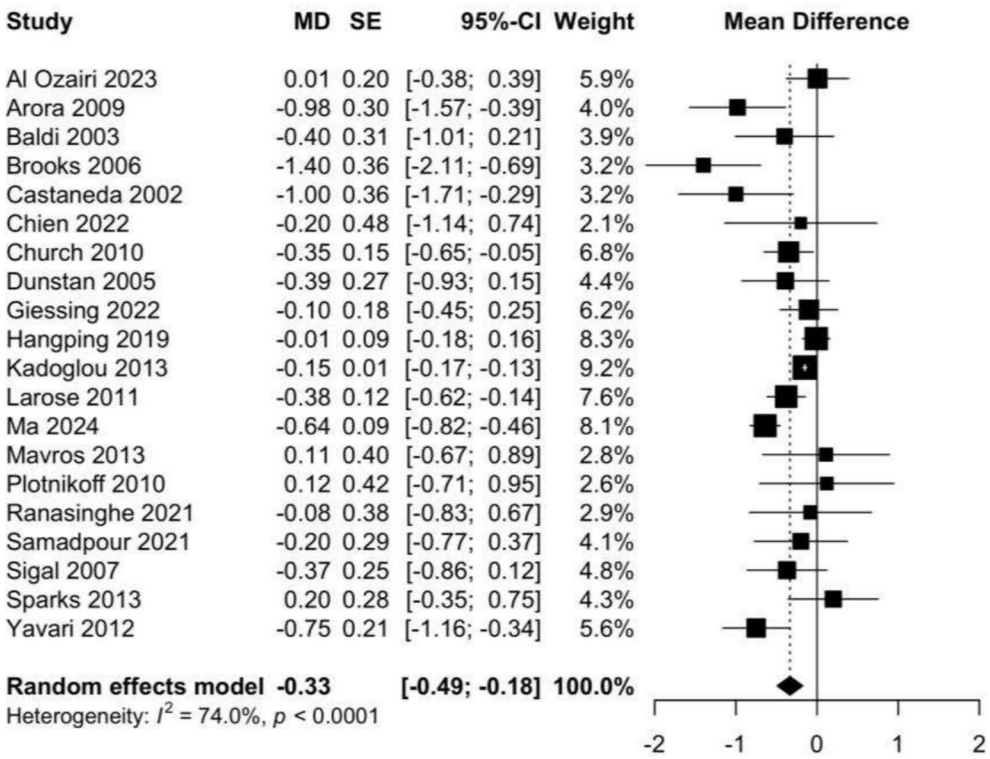

Fig. 2 All studies: Effects of resistance training versus control conditions

Quality of randomized controlled trials and certainty of evidence supporting recommendations for gym-based and home-based resistance training

The methodological quality score of the included studies, assessed using the PEDro scale, ranged from 3 to 8, with an average score of 6 ("good" quality). The studies' PEDro scores are presented in Table 1, with scoring details available in the appendix (Additional file 3).

The certainty of evidence for recommending resistance training in individuals with T2DM was evaluated using the GRADE approach. A half-level downgrade was applied for risk of bias, as the mean PEDro score of 6 suggests some uncertainty. An additional half-level downgrade was applied for inconsistency due to substantial heterogeneity (I^2: 74%). No downgrade was deemed necessary for indirectness, as the patient population, interventions, comparators and outcomes were fully aligned with our PICO scheme. No downgrade was applied for imprecision, as the observed mean reduction in HbA1c (−0.33%) is of moderate magnitude, with a CI that can be assumed to fall within the range of clinical relevance [52]. Furthermore, publication bias was not suspected (symmetric funnel plot). The two half-level downgrades yield an overall moderate certainty of evidence rate.

Similarly, in the gym-based subgroup, the certainty of evidence was downgraded by half a level for risk of bias (mean PEDro score: 5) and by another half-level for inconsistency (I^2: 79%), but not for indirectness or imprecision. There was also no significant publication bias. The certainty of evidence was therefore also rated as moderate.

In contrast, the home-based subgroup was downgraded by half a level for risk of bias (mean PEDro score: 6), but no downgrade was applied for inconsistency (I^2: 0%) or indirectness. However, a two-level downgrade was applied for imprecision due to the trivial effect size (mean HbA1c change: +0.12%) and a CI that includes both benefits and harms. No indication of publication bias was found. These downgrades resulted in an overall rating of very low certainty of evidence for recommending home-based resistance training.

Discussion

This meta-analysis offers a comprehensive evaluation of the effects of resistance training on glycemic control, highlighting the influence of gym- and home-based interventions.

Gym-based and home-based resistance training to improve glycemic control

The overall findings confirm a small but statistically significant positive effect of resistance training on

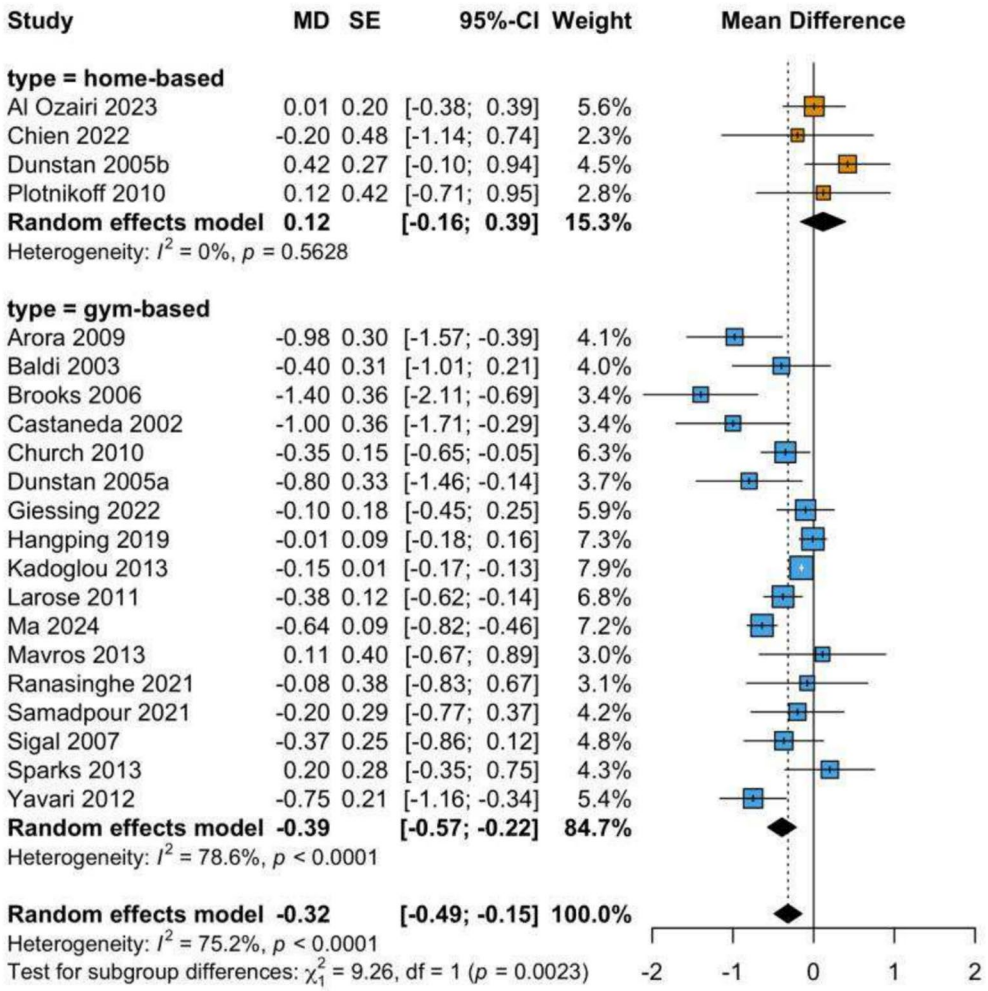

Fig. 3 Subgroup analysis: Effects of home-based or gym-based resistance training versus control conditions

HbA1c levels. One key distinction emerged between gym-based and home-based resistance training. While gym-based interventions demonstrated a statistically significant reduction in HbA1c, home-based training did not reveal a meaningful effect. The significant difference between these subgroups suggests that structured, supervised training in gym settings may be more effective in improving glycemic control compared to home-based programs. No significant moderating effect of intervention duration or total training volume was observed, indicating that other factors may contribute to the variability in HbA1c responses to resistance training. The subgroup analysis of different resistance training methods revealed differences in terms of effectiveness. Hypertrophy training resulted in a significant reduction in HbA1c, whereas strength endurance training and maximal strength training did not show significant effects. These findings highlight the importance of targeted gym-based training interventions to optimize metabolic outcomes in individuals with T2DM.

The results from our random-effects model demonstrated a significant effect of RT, with a small but significant pooled mean difference favoring RT vs control (MD = −0.33, 95% CI −0.49 to −0.18). These findings are consistent with previous meta-analyses reporting a small but significant positive effect of RT on HbA1c levels among patients with T2DM [9–14]. The meta-analysis conducted by Su et al. in 2023 [53] found a moderate effect of RT on HbA1c in T2DM individuals (SMD = −0.63, 95% CI −0.97 to −0.29) [51]. Su et al. [53] included a different sample of studies and used different inclusion/exclusion criteria. Notably, some studies did not specify training supervision [54, 55], failed to indicate whether patients with type 1 diabetes mellitus were excluded [55] or lacked a passive control group [56]. Our subgroup

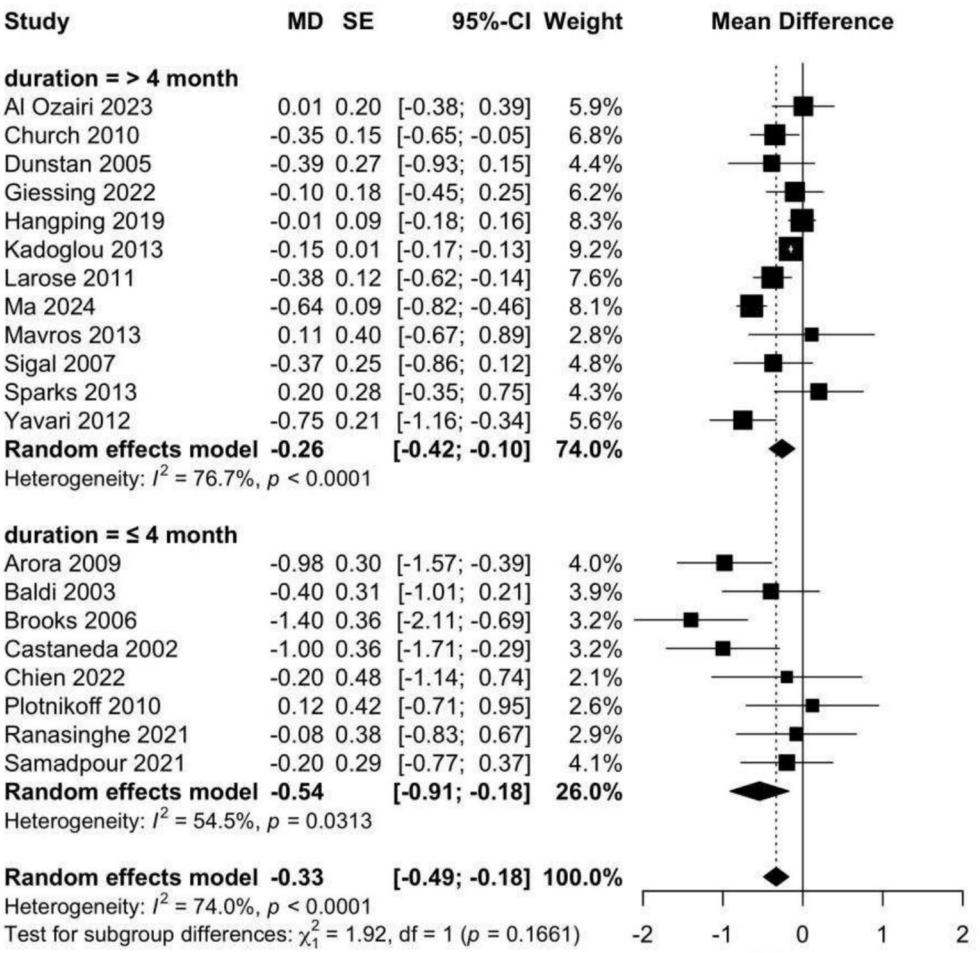

Fig. 4 Subgroup analysis: Effects of long-term (> 4 months) or short-term (≤ 4 months) resistance training versus control conditions

comparison suggests that supervised gym-based RT is more effective than home-based RT for improving HbA1c, with only gym-based RT showing a significant, albeit small, but clinically relevant reduction in HbA1c. Similar to our findings, a previous meta-analysis by Pan et al. in 2018 [57] revealed that only supervised RT led to significant reductions in HbA1c, while unsupervised RT showed no such effect. However, an important limitation of the aforementioned meta-analysis is the small number of studies included, with only five studies on supervised RT and two on unsupervised RT.

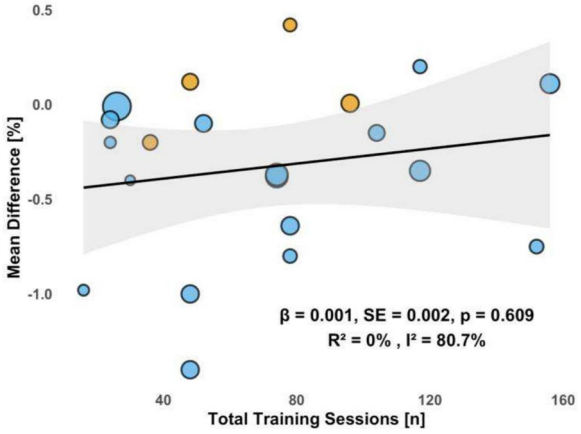

Fig. 5 Meta-regression analysis: Association between total number of training sessions and changes in HbA1c levels

What are the potential reasons for the limited effectiveness of home-based resistance training interventions? Exercise supervision has been shown to be an important factor in improving adherence to exercise programs [58–60]. The typical lack of supervision in home-based RT may contribute to lower adherence, which in turn could explain the lack of effect of home-based resistance training on HbA1c. Supporting this possible explanation, Dunstan et al. [40] reported lower adherence to their home-based RT intervention compared to their supervised gym-based RT intervention. Furthermore, the limited availability of equipment for home-based RT may

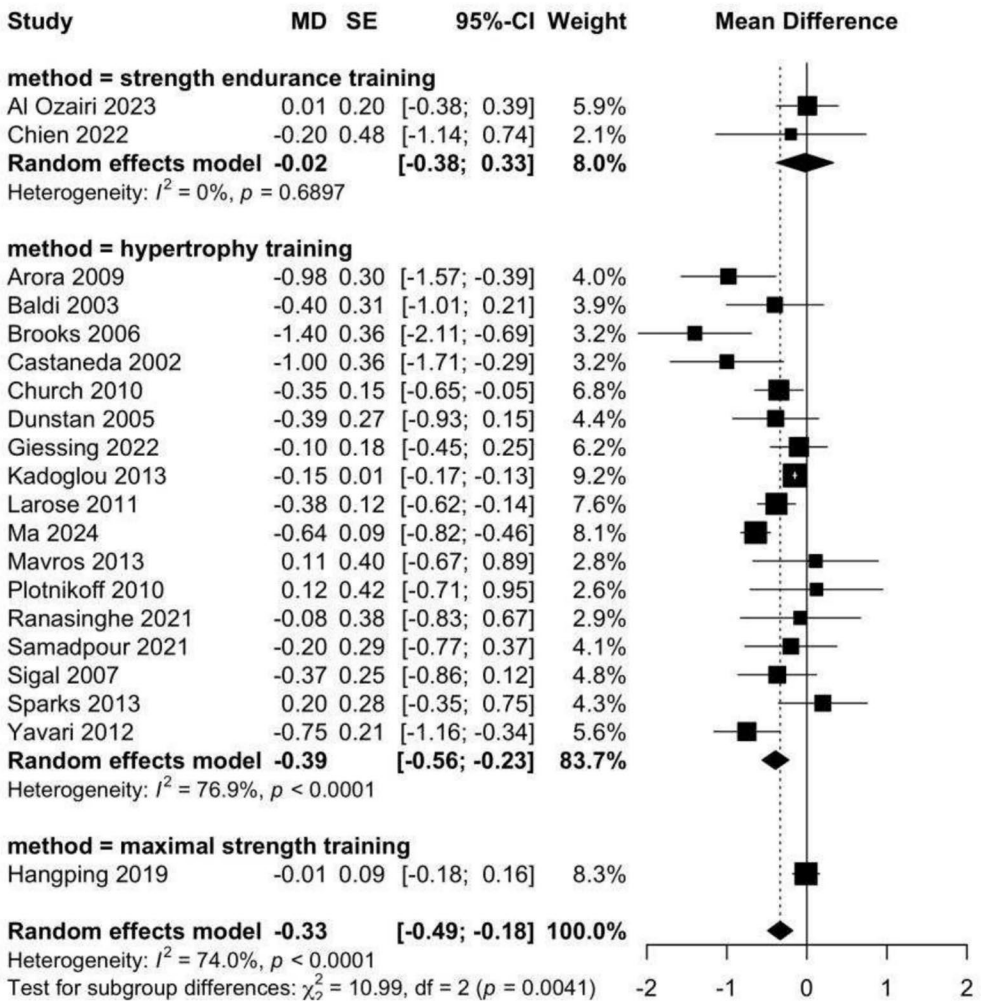

Fig. 6 Subgroup analysis: Effects of strength endurance, hypertrophy or maximal strength training versus control conditions

have hindered the ability to achieve the optimal intensity required to increase muscle mass and improve glycemic control. The equipment implemented in the home-based RT interventions analyzed in this meta-analysis included sandbags, dumbbells and ankle weights, with only one study incorporating a multigym apparatus. In this context, it may be more difficult to achieve precise load dosing with home training equipment than with equipment of a gym, where training is mainly carried out on training machines. Dunstan et al. [40] reported a lower intensity during the home-based RT compared to the gym-based RT intervention in their study (60–80% of 1RM, and 75–85 of 1RM%, respectively). There may also be more disruptive factors at home, such as unexpected visits from friends or interruptions from children, which can lead to frequent breaks or cancellations of training sessions, and thereby reduce overall efficiency. Future studies on home-based resistance training should record actual adherence to training protocols, e.g. through the use of wearables, and report the true intensities and volumes achieved. This can help identify the underlying causes of the inefficiency of home-based resistance training programs.

Moderating factors

It is possible that additional training-related factors may influence the effectiveness of resistance training in individuals with T2DM and could have influenced the meta-analysis' result in terms of the differences in observed effectiveness between gym- and home-based workouts.

Consistent with this assumption, a further subgroup analysis revealed a significant effect of the training method on HbA1c changes. Hypertrophy training showed a significant effect, whereas endurance strength training and maximal strength training did not. However, this finding should be carefully interpreted, as strength endurance training was used in only two studies and maximal strength training in only one, while the

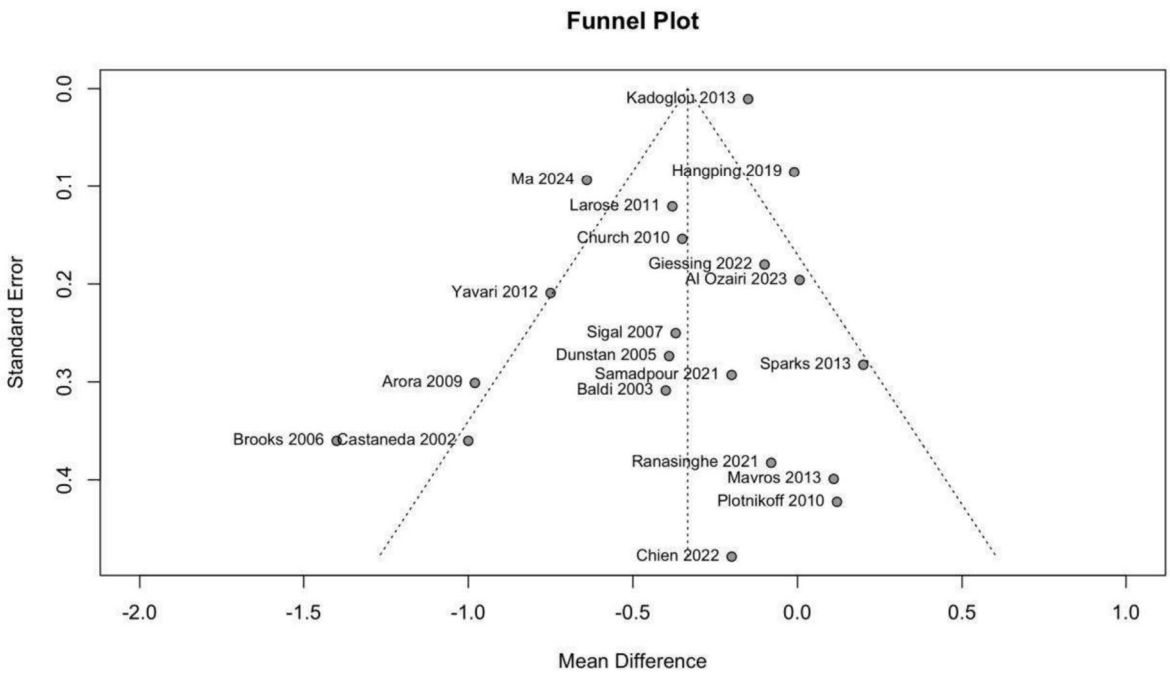

Fig. 7 Funnel plot to visually detect *publication bias*

majority of studies used hypertrophy training protocols. Notably, two of the four home-based RT interventions in our meta-analysis were classified as endurance strength training [34, 38]. The distribution between the two groups (home-based and gym-based resistance training) with reference to the training methods used was therefore unbalanced. The literature examining the impact of RT intensity on HbA1c levels has yielded conflicting results. The meta-regression analysis by Liu et al. [14] found correlations between RT intensity and changes in HbA1c, suggesting that high-intensity RT (75–100% of 1-RM) is more beneficial than low-to-moderate-intensity RT (20–75% of 1RM). In contrast, another meta-analysis by Lee et al. [13] examining the role of RT variables in elderly patients with T2DM did not find any significant effect of RT intensity on HbA1c changes.

Our subgroup analysis comparing the effects of short- vs long-term RT on glycemic control revealed no significant differences between the subgroups. Consistent with our findings, existing literature has not shown intervention duration to be a significant factor influencing HbA1c outcomes [11–13, 53]. This may be due to insufficient progression in training workload and/or lower adherence to the training program in the long term. The limited additional positive effects from longer intervention periods may also be due to the fact that HbA1c levels improve after a certain duration and become more difficult to improve further once they have neared the normal range. A lack of variation in appropriate training stimuli could also explain this result. A recent meta-analysis demonstrated that combined exercise and nutrition interventions can lead to significant reductions in HbA1c [61]. Integrating dietary interventions with RT may offer a promising strategy to provide additional stimuli to enhance glycemic control, and may also be useful in the long term [5, 6]. However, the duration of the training period is unlikely to have influenced the meta-analytic comparison of the effectiveness of gym-based versus home-based resistance training.

Limitations

This systematic review with meta-analysis has some limitations. Included studies demonstrated inconsistencies, e.g. in terms of the number and choice of exercises, equipment used, duration of a single session, training frequency per week or intervention duration, resulting in significant heterogeneity among the included trials. Furthermore, several influencing factors, such as medication use, the presence of other chronic diseases, or different warm-up/cool-down procedures (that were not always reported in detail in the included studies) were not considered in the present analyses.

This study focused on HbA1c as an outcome measure, although other outcomes (cardiovascular health, well-being) are also important. However, the study's strength is that many subgroup analyses were carried out and that the relevance of different training modalities for changes in the selected outcome was clarified. Despite

the subgroup and sensitivity analyses, substantial heterogeneity remained. This likely reflects variation in intervention characteristics, participant profiles, and study methodology. However, these aspects were inconsistently reported and addressed in only a limited number of studies, preventing further stratified analyses. Future primary research should systematically capture and report such variables to enhance the interpretability of pooled effects.

Conclusions

In conclusion, this meta-analysis confirms the effectiveness of RT in reducing HbA1c levels in individuals with T2DM. Notably, our findings indicate that supervised gym-based RT is more effective than home-based RT in reducing HbA1c levels. The certainty of evidence for recommending resistance training in individuals with T2DM was rated as moderate for both resistance training in general and gym-based resistance training, but very low for home-based resistance training. The present findings offer valuable insights for refining RT protocols to enhance glycemic control in individuals with T2DM.

Abbreviations

HbA1c	Glycated hemoglobin
MD	Mean difference
PICO scheme	Patient-Intervention-Comparison-Outcome scheme
RCT	Randomized controlled trial
RM	Repetition maximum
RT	Resistance training
SD	Standard deviation
SE	Standard error of the mean
SMD	Standardized mean difference
T2DM	Type 2 diabetes mellitus

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

Supplementary Material 3.

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Author contributions

Conceptualization: C.B.; Literature search: M.B., S.G.; Quality assessment: M.B./S.G., V.IBP., C.B.; Data curation and analysis: M.B., V.IBP., S.H., C.B.; Methodology: S.H., C.B.; Supervision: C.B.; Validation: C.B., S.H.; Visualization: S.H., C.B.; Writing: original draft: V.IBP., C.B.; Review & editing: S.H., C.B. All authors read and approved the final manuscript.

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Availability of data and materials

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

C.B. received honoraria from Abbott, Lilly and Novo Nordisk and is a member of the Abbott Advisory Board. The companies had no role in the manuscript's preparation or decision to publish it. All other authors declare that they have no conflict of interest.

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